Patient Voice Assistant Workflow

Hackathon 24-Hour Build Plan — Version 3

The Pitch

> **"A voice assistant that books care, and when it's an emergency, it puts the patient first, automatically."**

The emergency flow with slot pre-emption is your differentiator. Every other booking bot handles normal flow. You handle the edge case that matters most.

Build Exactly This (Priority Order)

1. Realtime Voice Session ✓ Core

- Higgs Realtime WebSocket (wss://api.boson.ai/v1/realtime)
- Mic capture + voice playback (24 kHz PCM16)
- Server-side VAD turn detection
- **Barge-in support** (interrupt playback when patient speaks)
- Live captions from response.output_audio_transcript.delta
- **Keyword highlighting** in captions:
  - Doctor names & times → **Blue**
  - Payment words → **Amber**
  - Emergency words → **Red** (e.g., "call emergency services", "severe")

2. Normal Booking Flow (End-to-End) ✓ Core

- Doctor suggestions (menu of 5 doctors)
- Available time slots
- **5-minute confirmation hold**

- Mock payment page (Done button)
- Receipt SMS with booking ID
- Single prompt rule: "describe problem, then book"
- Keyword map for symptom detection

3. Emergency Flow ✓ Differentiator

- **Red-flag detection** on live transcript
- Capture: name, address, phone
- Create same-day slot with **EMERGENCY tag**
- Show displaced normal booking on dashboard (visual proof of pre-emption)
- Alert state on character

4. Browser Animated Character ✓ UI

- **Four states:** idle, listening, speaking, alert
- **Four cards below:** doctor, time, booking ID, payment status
- Simple SVG or CSS animations
- Updates in real-time as conversation flows

5. Virtual Phone Widget ✓ UI

- Displays incoming SMS messages

- Mock SMS: booking confirmations + IDs
- Shows 2-3 most recent messages

6. Avatar Clips (Pre-Rendered) ✓ Boson API requirement

- **3 clips,** cached/reusable:
  - Welcome: "Hi, I'm your booking assistant..."
  - Emergency reassurance: "We're getting you help immediately..."
  - Booking confirmed: "Your appointment is booked for [time]..."
- 2-3 seconds each, Higgs Avatar API
- Plays at key moments (not every turn)

7. STT Safety Net ✓ Reliability

- Keyword check on live transcript
- If model misses red flags, safety net catches them
- Example: patient says "I can't breathe" → force alert state even if model didn't flag it
- Override confidence with keyword heuristics

8. Demo-Mode Buttons ✓ Fallback

- Canned phrases: "I have chest pain", "Book me a normal appointment", "It's urgent"
- One-click replay
- **Recorded backup video** (30 sec, worst-case scenario)

Remove or Postpone

Delete (Out of Scope):

- OTP / PIN verification
- Cancellation flow
- Human hand-off ("talk to an agent")
- Reconnect / session resumption logic
- Stripe integration (use mock payment)
- Real Twilio SMS (mock only)
- Recap strip / detailed call history

Minimize:

- Extra character states (no "waiting", "celebrate", etc.)
- Doctor database depth (stick to 5 doctors)

- Avatar speaking full conversation turns (pre-rendered clips only)
- Web search / external lookups

Only if Time Allows (Hour 16+):

- Reschedule logic (~30 min)
- Check appointment status (~15 min)
- *Do NOT attempt:* full CRM, provider integrations, real payment

---

Demo Sequence (Rehearsed)

1. Emergency First (1-2 min)

- **Patient:** "I have severe chest pain and shortness of breath"
- **System:** detects red flags, captures details
- **Screen:** character goes red (alert), booking shows EMERGENCY tag
- Show displaced normal booking sliding down

2. Normal Booking (1-2 min)

- **Patient:** "I'd like to book a routine checkup"
- **System:** suggests doctor, time, confirms payment
- **Screen:** character smiling, cards filling in
- SMS arrives with confirmation

3. Barge-in Demo (30 sec)

- While avatar is speaking, patient interrupts
- Audio stops, system immediately switches to listening

Rehearse this 2x perfectly before demo time.

The 5 Rules That Decide Winning

Rule	Impact	How to Track
Complete booking by hour 8	Entire second half for polish & demo prep	Set timer; if not done, cut features now

Freeze features by hour 18	Buffer for unexpected issues, sleep before demo	Hard stop; no new code after this
Rehearse demo until smooth	Judges see polish, not bugs	Run 2x perfectly in a row minimum
Demo order matters	Emergency first → judges see your edge case	Brain them early with differentiator
Have cost-per-call ready	Shows you've thought about margins	~$0.05–$0.10 per 5-min call (Boson docs)

Technical Architecture

API Endpoints (Proxy Routes)

```
POST /api/realtime/client-secret → POST /v1/realtime/client_secrets WebSocket /v1/realtime (direct from browser, ephemeral key) POST /api/videos → POST /v1/videos (Avatar clips) POST
```

```
/api/tts/speech → POST /v1/audio/speech (if needed)
```

Data Flow

1. **Backend** (Node.js/Python): holds long-lived BOSON_API_KEY
2. **Client-Secret Mint:** browser requests → backend creates ephemeral key → browser opens WebSocket
3. **Realtime Session:** browser ↔ Higgs API (24 kHz PCM16, VAD, transcript streaming)
4. **Avatar Jobs:** browser requests → backend submits → polls status → downloads MP4
5. **Mock Services:** SQLite or in-memory for doctors, bookings, SMS log

Key Boson APIs in Use

- **Higgs Realtime** (primary conversation)
- **Higgs Avatar** (video clips)
- **Higgs STT 3.1** (transcription, inside Realtime)
- **Higgs TTS 3** (voice in Avatar, optionally in Realtime for fallback)

24-Hour Timeline

Phase	Hours	Allocation	Milestone
Setup & Boilerplate	0–1	Backend scaffold, Boson SDK setup	Running /api/health ✓
Realtime Integration	1–4	WebSocket, mic/speaker, VAD, captions	Live voice session works ✓
Normal Booking	4–7	Prompt logic, slot selection, payment mock	E2E normal flow works ✓
Emergency Detection	7–9	Red-flag keyword check, override logic	Emergency path works ✓
UI & Polish	9–15	Character, cards, phone widget, Avatar clips	Looks polished ✓
		Fallback	

Demo Prep & Rehearsal	15–22	buttons, backup video, script, 2x dry-runs	Demo is smooth ✓
Buffer / Fixes	22–24	Unexpected bugs, final tweaks	Ship ready ✓

Team Allocation (Example 3-person)

Person	Strength	Primary Work	Secondary
A	Backend/APIs	Realtime setup, client-secret minting, database mock	Server architecture
		Character,	

B	Frontend/React	cards, captions, phone widget	UI polish
C	Product/Demo	Prompt tuning, emergency logic, demo script	Backup video recording

(Adjust if team is 2 or 4 people; parallelism is your friend.)

Honest Odds

This version gives you the **best chance to win** because:

- **Differentiator is clear:** emergency pre-emption (no other team will have this)
- **Scope is realistic:** fits 24 hours, avoids feature creep
- **Demo is tight:** rehearsed, dramatic (emergency flow first), shows polish
- **Boson APIs fully exercised:** TTS, Voices (maybe),

Avatar, Realtime

What you DON'T want to explain to judges:

- "We ran out of time and didn't finish the normal flow"
- "The demo crashed mid-booking"
- "We built 10 features, 3 of them barely work"

What you DO want to explain:

- "We focused on the edge case everyone ignores: when a patient needs emergency care, our system pre-empts their normal booking"
- "Here's the live voice conversation, caption highlighting, and the avatar reassuring them"
- "It's a complete, working system that judges can use right now"

Final Checklist

- [ ] Boson API key in .env
- [ ] Backend scaffold (Node/Python)
- [ ] WebSocket connection to Higgs Realtime
- [ ] Mic + speaker plumbing (PCM16, 24 kHz)
- [ ] Transcript streaming + caption display

- [ ] Keyword color coding (red/amber/blue)
- [ ] Doctor/slot data (mock database)
- [ ] Red-flag detection (emergency trigger)
- [ ] Character state machine (idle → listening → speaking → alert)
- [ ] Card UI (doctor, time, booking ID, payment)
- [ ] Virtual phone widget + mock SMS
- [ ] Avatar clips 3x (pre-rendered)
- [ ] Demo-mode canned phrases + video fallback
- [ ] Prompt rule for normal flow
- [ ] 2x full-flow demo rehearsal
- [ ] Backup video (just in case)

Pro Tips

1. **Prioritize the emergency flow.** It's your story. Get it working early.
2. **Use mock services everywhere.** Real Twilio,

Stripe, external APIs = risk. SQLite + mock = reliability.

3. **Record a backup video NOW.** If live demo fails, you still have proof of concept.
4. **Test barge-in aggressively.** Interrupt the avatar mid-sentence. Show it works.
5. **Caption highlighting is a quick win.** Spend 30 min on it; judges love it.
6. **Keep the character simple.** A smiley face changing to an alert triangle beats a 3D avatar. CSS > complexity.

You've Got This

Goal: Ship a working, polished, differentiated demo in 24 hours.
Edge: Emergency pre-emption + real-time voice + smart captions.
Win condition: Judges say "I've never seen a booking bot handle emergencies like this."
Rehearse until the demo is flawless. Ignore

> **edge cases not in the flow. Sleep the night before.**

Generated for Hackathon 24-Hour Sprint
Boson AI (Higgs) APIs: TTS 3, Voices, Avatar, Realtime
Version 3 — Final